THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Using gcd$(212, 84) = 4$, the LCM of 212 and 84 is: 1

(a) 4448 **(b)** 4452 **(c)** 4456 **(d)** 4664

2. To test whether 1427 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{1427}\rfloor$. The last number checked is: 1

(a) 37 **(b)** 38 **(c)** 713 **(d)** 36

3. A counter starts at 0. For i = 1, 2, …, 60, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 28 **(b)** 30 **(c)** 27 **(d)** 29

4. By Euclid's algorithm, gcd$(2851, 2230)$ is: 1

(a) 0 **(b)** 2 **(c)** 3 **(d)** 1

5. Start with $n = 38$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 23 **(b)** 21 **(c)** 20 **(d)** 22

6. Using Euclid's subtraction algorithm for gcd$(1962, 1044)$, the number of subtraction steps needed is: 1

(a) 13 **(b)** 14 **(c)** 16 **(d)** 15

7. Using the division form of Euclid's algorithm for gcd$(2951, 2455)$, the number of divisions needed is: 1

(a) 9 **(b)** 6 **(c)** 8 **(d)** 7

8. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 8759, the final r is: 1

(a) 9668 **(b)** 9587 **(c)** 9578 **(d)** 9569

9. Using gcd$(65, 285) = 5$, the LCM of 65 and 285 is: 1

(a) 3700 **(b)** 3705 **(c)** 3710 **(d)** 3770

10. To test whether 1697 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{1697}\rfloor$. The last number checked is: 1

(a) 42 **(b)** 41 **(c)** 848 **(d)** 40

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the gcd of 559 and 598 by the division form of Euclid's algorithm and use it to find their LCM. 2

12. Find the gcd of 452, 522 and 158 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 2

13. A floor of 468 cm by 512 cm is to be covered by identical square tiles, as large as possible, with no cutting. Use Euclid's algorithm to find the tile side and the number of tiles. 2

14. Find the gcd of 375 and 627 by the division form of Euclid's algorithm and use it to find their LCM. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Trace: count = 0; for i = 1 to 8: for j = i to 8: count = count + 1. Find the final count and explain the pattern. 3

16. Trace: total = 0; for i = 1 to 7: if i is a multiple of 4, add i^2 to total. List the numbers added and the final total. 3

17. Start with n = 30. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

OR

Trace: count = 0; for i = 1 to 9: for j = i to 9: count = count + 1. Find the final count and explain the pattern. 3

18. Trace: total = 0; for i = 1 to 10: if i is a multiple of 3, add i^2 to total. List the numbers added and the final total. 3

OR

Start with n = 17. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the largest number that divides both 2120 and 689 exactly, using the division algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 2712, 1554 and 2932 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 5

20. Find the largest number that divides both 1813 and 623 exactly, using the division algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 1372, 1154 and 1080 by applying Euclid's division algorithm twice: first to two numbers and then to the result and the third. 5

END OF QUESTION PAPER — Total: 40 Marks